

One Pic is All it Takes: Poisoning Visual Document Retrieval Augmented Generation with a Single Image (Short Paper)

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Abstract

Multimodal retrieval augmented generation (M-RAG) has recently emerged as a method to inhibit hallucinations of large multimodal models (LMMs) through a factual knowledge base (KB). However, M-RAG also introduces new attack vectors for adversaries that aim to disrupt the system by injecting malicious entries into the KB. In this work, we present a poisoning attack against M-RAG targeting visual document retrieval applications, where the KB contains images of document pages. Our objective is to craft a single image that is retrieved for a variety of different user queries, and consistently influences the output produced by the generative model, thus creating a universal denial-of-service (DoS) attack against the M-RAG system. We demonstrate that while our attack is effective against a diverse range of widely-used, state-of-the-art retrievers (embedding models) and generators (LMMs), it can also be ineffective against robust embedding models. Our attack not only highlights the vulnerability of M-RAG pipelines to poisoning attacks, but also sheds light on a fundamental weakness that potentially hinders their performance even in benign settings.

1 Introduction

Large multimodal models (LMMs) enable a wide range of zero-shot capabilities spanning a variety of input modalities, such as language, vision, and audio (Bordes et al., 2024; Liu et al., 2023). However, similar to unimodal large language models (LLMs), they can suffer from hallucinations, generating outputs that are not grounded in factual information (Huang et al., 2025). Retrieval augmented generation (RAG) (Lewis et al., 2020) attempts to address this issue by retrieving trusted information chunks, that are relevant to the user query, directly from a knowledge base (KB). The retrieved information is then included in the input to the generative model to inform its generation.

The effectiveness of RAG relies primarily on the trustworthiness of the information in the KB. Challenging this assumption, recent work has shown that textual RAG pipelines are vulnerable to poisoning attacks, where an attacker injects malicious information into the KB (Zou et al., 2024; Xue et al., 2024). To create an impactful attack, the injected information must simultaneously (1) have a high chance of being retrieved, and (2) influence the output of the generative model. These attacks have recently been extended to multimodal RAG (M-RAG) pipelines (Chen et al., 2022; Lin and Byrne, 2022) by concurrent research (Ha et al., 2025; Liu et al., 2025).

Both of those concurrent works consider that the user queries and the KB consist of text-image pairs, thus the text modality is common between queries, KB items, and LMM outputs. This shared modality greatly simplifies the attack procedure. For instance, an adversary could simply inject the expected user request alongside the malicious response into the KB (Liu et al., 2025). This trivial—but effective—attack is not possible in many practical use cases, such as visual document retrieval (Faysse et al., 2024; Yu et al., 2024) where the KB consists purely of images (document pages), while both the user queries and the LMM outputs are textual. Therefore, it remains unclear whether M-RAG poisoning would still be possible in this practical setting.

In this paper, we aim to bridge this gap in the literature by investigating poisoning attacks on visual document RAG (VD-RAG). We adopt a gradient-based adversarial approach (Goodfellow et al., 2014; Madry et al., 2017) to craft an image that is retrieved, and consistently influences the generation of every possible user query, thus leading to a universal denial-of-service (DoS) attack against the RAG pipeline. In particular, we make the following contributions: (1) We propose the first poisoning attack against VD-RAG systems.

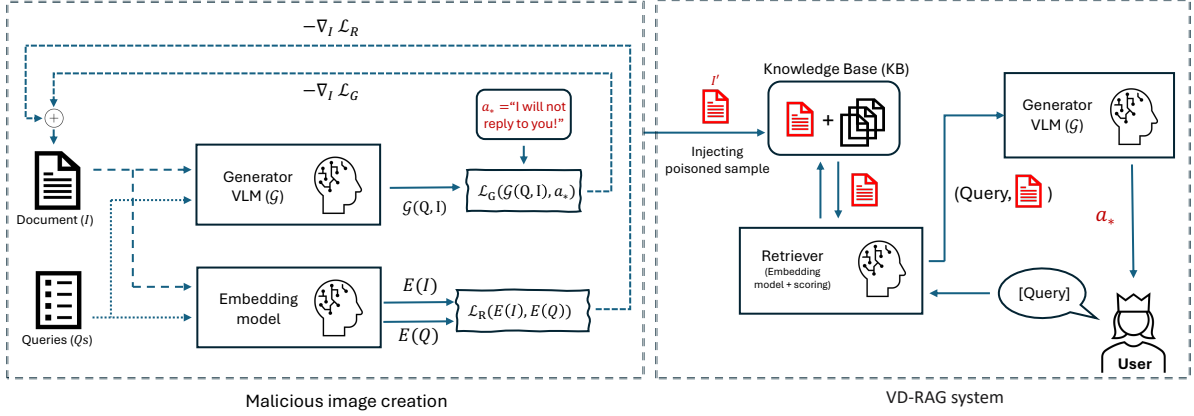


Figure 1: Overview of our attack. We select an arbitrary document/image I and optimize it against all queries Q in the training set (shown on the left). The resulting poisoned document I' is then injected into the KB. When the attack is successful, I' is retrieved and causes the generator $\mathcal{G}$ to malfunction (shown on the right).

(2) We demonstrate that our multi-objective optimization approach allows an adversary to craft a single image that causes DoS against the entire RAG pipeline. (3) We investigate several factors of attack success, such as the embedding model employed, the LMM, and the stealthiness of the injected image.

2 VD-RAG Pipeline

A visual document RAG pipeline consists of three main components. First, a *knowledge base* (KB) $\mathcal{K} = \{I_1, \dots, I_K\}$ containing a set of K images, each corresponding to a page in a document. Second, a *retriever* $\mathcal{R}$ uses a multimodal embedding model $E(\cdot)$ that projects user queries (text) and KB images into a common vector space. The retriever then computes a similarity score $S(E(q), E(I))$ between a user query q and each image in $I \in \mathcal{K}$. Common similarity metrics for RAG retrievers include cosine similarity and *MaxSim* proposed in (Faysse et al., 2024). For each user query q , the retriever retrieves the top- k relevant items from $\mathcal{K}$ according to the similarity score, where $k \ll |\mathcal{K}|$. In other words, the retriever computes $R(q, \mathcal{K}) = \text{top-}k_{I \in \mathcal{K}} S(E(q), E(I))$. The third component is a *generator* $\mathcal{G}$ which is a vision language model (VLM) used to generate a response g to the user’s query q with the retrieved images in its context window. That is, $g = \mathcal{G}(q, R(q, \mathcal{K}))$.

3 Poisoning Attacks against VD-RAG

Threat Model. We consider an attacker that aims to disrupt the operation of the visual document RAG system by causing the retriever to retrieve irrelevant images and the generator to generate

unhelpful responses to user queries. To achieve this goal, the attacker is capable of injecting documents/images into the KB. In our work, we assume a weak attacker that can inject only one malicious image I' into the KB s.t. $\mathcal{K}' = \mathcal{K} \cup I'$. In order for this malicious image to cause disruption to the RAG pipeline, this image has to be crafted to be retrieved often, and when retrieved, it has to cause the VLM to generate unhelpful responses.

We examine both *white-box* and *black-box* settings. In the white-box setting, the attacker is assumed to have knowledge of both the embedding model and the VLM. This is motivated by the vast amount of competent open-source text and multimodal (Faysse et al., 2024) embedding models, as well as VLMs (Duan et al., 2024). In the black-box setting, we consider that the attacker uses other open-source models than those used in the VD-RAG system to craft the malicious image, and we consequently evaluate the transferability between adversarial samples from one model to another.

Proposed Attack. Building upon the work of (Zou et al., 2024), a successful RAG poisoning attack must meet two conditions (modified for our goal of a universal untargeted attack). First, the *retrieval condition* requires that the malicious image is retrieved as often as possible. Second, the *generation condition* requires that when present in the context window, the malicious image must cause the generator to generate a specific response. In order to compute a malicious image I' that meets the above two conditions, we adopt the framework of gradient-based adversarial examples, initially proposed against neural network based image clas-

sifiers (Goodfellow et al., 2014; Carlini and Wagner, 2017; Madry et al., 2017). In particular we jointly optimize the image to minimize a loss function $\mathcal{L}_{RAG}$ capturing both a retrieval loss $\mathcal{L}_R$ and a generation loss $\mathcal{L}_G$ as follows

$$\mathcal{L}_{RAG} = \lambda_R \mathcal{L}_R + \lambda_G \mathcal{L}_G, \quad (1)$$

where λ_R, λ_G are coefficients,

$$\mathcal{L}_R = \sum_{i=1}^{|Q|} 1 - S(q_i, I') \quad (2)$$

$$\mathcal{L}_G = \sum_{i=1}^{|Q|} CE(\mathcal{G}(q_i, R(q_i, \mathcal{K}')), a_*), \quad (3)$$

where Q is the set of user queries used to train the attack, $CE(\cdot)$ is the cross entropy loss, a_* is the desired target answer by the attacker

4 Experiment Design

Datasets. We evaluate our attacks on two visual document retrieval datasets taken from the ViDoRe benchmark versions 1 and 2. That is, we use the ViDoRe-V1-AI and ViDoRe-V2-ESG datasets (Faysse et al., 2024). ViDoRe-V1-AI consists of 100 queries and 1000 images (with exactly one relevant ground-truth image in the KB per query), while ViDoRe-V2-ESG consists of 52 queries and 1538 images with an average of 2.5 relevant images per query¹. We split each dataset into a training set (80%, used to optimize the malicious image) and a test set (20%).

Embedding Models. We use a mix of embedding models that range in their sizes, recency, and target applications: First, CLIP-ViT-LARGE is a seminal multimodal 0.4B parameter model proposed in (Radford et al., 2021) trained using contrastive learning to achieve zero-shot image classification. Second, Jina-CLIP-v2 is a recent 0.9B parameter extension of CLIP (Koukounas et al., 2024) that is trained with additional multilingual text-image and text-text contrastive losses to bridge the modality gap problem (Liang et al., 2022) exhibited by CLIP. Finally, ColPali-v1.3 is a 3B parameter state-of-the-art model (Faysse et al., 2024) in visual document retrieval, using ColBERT-style (Khattab and Zaharia, 2020) late embedding interaction, and incorporating the *MaxSim* retrieval similarity metric. For the above models, we assume the retriever only retrieves the top-1 relevant image from the KB.

¹Not all images in the KB have to be relevant.

VLMs. We evaluate our attacks on two VLMs: SmolVLM-Instruct (2.2B) (Marafioti et al., 2025) and Qwen2.5-VL-3B-Instruct (3.75B) (Wang et al., 2024).

Evaluation Metrics. We evaluate the RAG system and the attack using the following performance metrics: 1.) Recall-B is the fraction of queries for which $\mathcal{R}$ retrieves a relevant image prior to attack. 2.) Recall-A is the fraction of queries for which $\mathcal{R}$ retrieves a relevant image after the attack. 3.) ASR-R is the fraction of queries for which the malicious image is retrieved by $\mathcal{R}$. 4.) ASR-G-Hard is the fraction of queries for which the VLM outputs the target response a_* verbatim, assuming that the malicious image is retrieved. 5.) ASR-G-Soft is the average embedding similarity between the VLM generated response and the target response a_* , assuming that the malicious image is retrieved, computed using the Jina-Embeddings-V3 (Sturua et al., 2024) text embedding model.

Attack Hyper-parameters. For each considered dataset, we pick an arbitrary image that is not relevant to any query, to act as a starting point for our optimization process. We produced our attacks through projected gradient descent (PGD) (Madry et al., 2017) using a linear learning rate schedule from 3×10^{-3} to 3×10^{-4} over 500 gradient steps with a batch size of 8 user queries, $\lambda_R = 2$, $\lambda_G = 1$, and a maximum image perturbation $\alpha = \frac{8}{255}$. The retrieval loss uses the cosine similarity between embeddings as $S(q, I)$, while the generation loss uses the target malicious reply: I will not reply to you!.

5 Results

Effect of Perturbation Intensity. Figure 2 shows how the maximum adversarial perturbation α affects attack success for a VD-RAG system consisting of CLIP-ViT-LARGE and SmolVLM-Instruct. We observe that attacks can almost perfectly satisfy both the retrieval and the generation conditions starting from $\alpha = \frac{8}{255}$. Therefore, in the rest of the paper, we consider only attacks with $\alpha = \frac{8}{255}$. In Appendix A, we provide visual examples of the stealthiness of an attack with $\alpha = \frac{8}{255}$. The figure also shows very little difference between the performance on the training and test sets, demonstrating that the malicious image does not overfit to the training dataset..

Comparison of Different Models. Table 1 presents an extensive evaluation of our attack

Table 1: Comparison of the attack impact on different embedding models and VLMs with regards to generation. The attack shows high values for ASR-G-Hard and ASR-G-Soft across all combinations and high ASR-R against Jina-CLIP-v2 and CLIP-ViT-LARGE, while ColPali-v1.3 exhibits good adversarial robustness.

Settings			Metrics				
Dataset	Embedder	VLM	Recall-B	Recall-A	ASR-R	ASR-G-Hard	ASR-G-Soft
ViDoRe-V1-AI	CLIP-L	SmolVLM	0.210	0.000	1.000	1.000	1.000
		Qwen2.5-3B	0.210	0.010	1.000	1.000	1.000
	Jina-CLIP	SmolVLM	0.000	0.000	0.750	0.900	0.951
		Qwen2.5-3B	0.000	0.000	0.750	0.950	0.987
	ColPali	SmolVLM	0.660	0.650	0.000	0.900	0.946
		Qwen2.5-3B	0.660	0.650	0.000	1.000	1.000
ViDoRe-V2-ESG	CLIP-L	SmolVLM	0.134	0.018	0.818	1.000	1.000
		Qwen2.5-3B	0.134	0.018	0.818	1.000	1.000
	Jina-CLIP	SmolVLM	0.000	0.000	1.000	1.000	1.000
		Qwen2.5-3B	0.000	0.000	1.000	0.818	0.991
	ColPali	SmolVLM	0.519	0.460	0.091	1.000	1.000
		Qwen2.5-3B	0.519	0.499	0.091	1.000	1.000

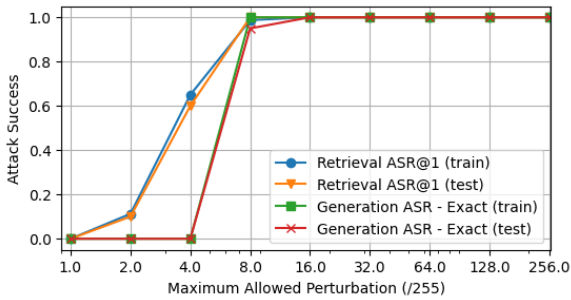


Figure 2: Attack success rate as a function of the perturbation intensity α .

against the datasets, the embedding models, and the VLMs considered. As the previous results indicate, we observe high values of ASR-R, ASR-G-Hard, and ASR-G-Soft for most of the considered models, except ColPali-v1.3 which is almost unaffected by our attack. Increasing the allowed perturbation to $\alpha = \frac{128}{255}$ and training over 1500 gradient steps did not produce better results against ColPali-v1.3. Otherwise, CLIP-ViT-LARGE and Jina-CLIP-v2 almost always retrieve our malicious image. In Appendix B we provide t-SNE visualizations (Van der Maaten and Hinton, 2008) of the embedding space that sheds light into the reasons for the superior robustness of ColPali-v1.3. Regarding generation, our attack consistently causes SmolVLM-Instruct and Qwen2.5-VL-3B-Instruct to generate the target answer *verbatim* for almost all user queries in the test dataset. We also observe an interesting interaction where Jina-CLIP-v2 seems to somewhat

inhibit the learning of effective attacks against generation. In general, the results show that our attacks are successful against a broad range of models.

Attack Transferability. While the above evaluations assumed white-box access to the attacker, we also evaluated our attacks in the black-box setting, where the adversary does not have access to the embedding model and VLM used in the RAG system. We did not observe any transferability of the attack across the different embedding and VLM models. Attacks trained against a certain combination (embedding model, VLM) only work against the same combination. Therefore, exploring attacks that are trained simultaneously against multiple models (Hu et al., 2025) is a direction for future research.

6 Conclusion

In this paper, we demonstrated the first poisoning attack against VD-RAG systems. Our attack showcased that conventional embedding models and VLMs are vulnerable to adversarial perturbation, where a single injected image is capable of causing a DoS of the entire RAG pipeline (both retrieval and generation). We also observed notable adversarial robustness of the ColPali-v1.3 embedding model and provided hints into why this is the case. Nevertheless, our attack provides the first step towards understanding the vulnerability of multimodal RAG systems and enhancing their robustness for future adoption.

Limitations

We do not consider defenses to our proposed attack; however, we plan to evaluate our attack against a variety of defenses including re-rankers, multimodal LLM-as-a-Judge (Chen et al., 2024), and poisoned sample detectors in future work. Additionally, we aim to examine whether adaptive attacks could be crafted against specific defenses. Our attacks can be detected by filtering images that are retrieved too frequently; however, our attack uncovers a fundamental weakness of M-RAG pipelines that could be useful for hardening through adversarial training. Our evaluation is not comprehensive, as we do not consider a large number of embedding models, VLMs, datasets, metrics, and transferability evaluations. In future work, we aim to provide an extensive evaluation of the attack showing the generalization of the results observed in this paper.

Ethical Consideration

The authors acknowledge the potential for misuse of this work through the creation of malicious inputs to AI systems. However, the authors believe that VD-RAG is a developing technology and, therefore, assessing the robustness of proposed approaches is critical to evaluating risks and mitigating them. Therefore, we hope that the results presented in this work will aid in the development of defenses for and the safe design of future VD-RAG systems.

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A Qualitative Attack Examples

We present a qualitative demonstration of our attack against the CLIP-ViT-LARGE embedding model and SmolVLM-Instruct VLM in Figure 3. Despite the success of the attack, the perturbed image is almost indistinguishable from the original.

B Embedding Space Visualizations

In this section we present 2D t-SNE visualizations (Van der Maaten and Hinton, 2008) of the embeddings of images and user queries using CLIP-ViT-LARGE, Jina-CLIP-v2, and ColPali-v1.3 in Figure 5 and Figure 6. The figures show that the image and textual embeddings produced by CLIP-ViT-LARGE and Jina-CLIP-v2 are easily separable (and thus easier to attack), while those computed by ColPali-v1.3 are harder to separate. These figures provide clues as to why it is harder to craft attacks against ColPali-v1.3 (i.e., it is harder to create one image that is closer to all query embeddings than all other image embeddings).



Figure 3: An example of a benign image from the ViDoRe-V1-AI Dataset (left) and its adversarially perturbed counterpart (right). Attack against CLIP-ViT-LARGE, SmolVLM-Instruct, with perturbation intensity $\alpha = \frac{8}{255}$. Result: ASR-R=1, ASR-G-Hard=1.

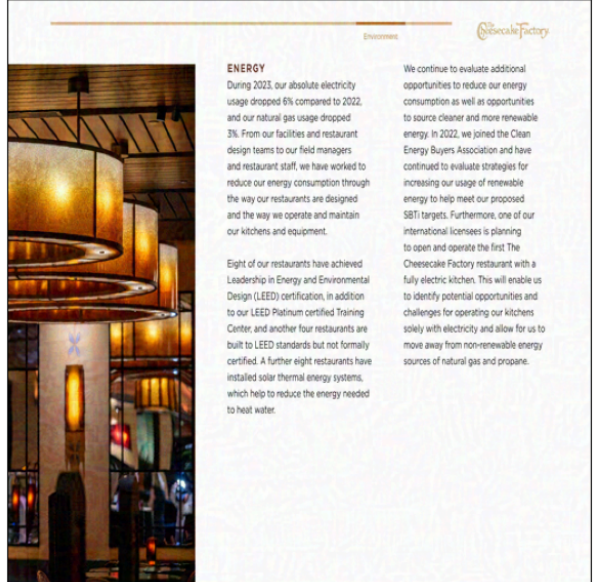
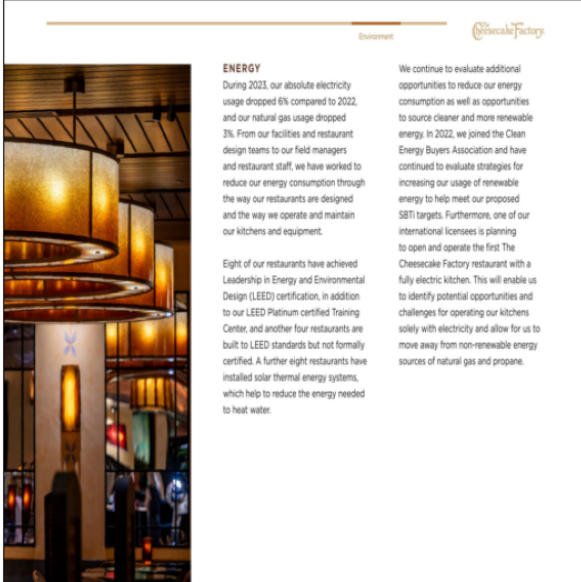


Figure 4: An example of a benign image from the ViDoRe-V2-ESG Dataset (left) and its adversarially perturbed counterpart (right). Attack against CLIP-ViT-LARGE, SmolVLM-Instruct, with perturbation intensity $\alpha = \frac{8}{255}$. Result: ASR-R=0.82, ASR-G-Hard=1.

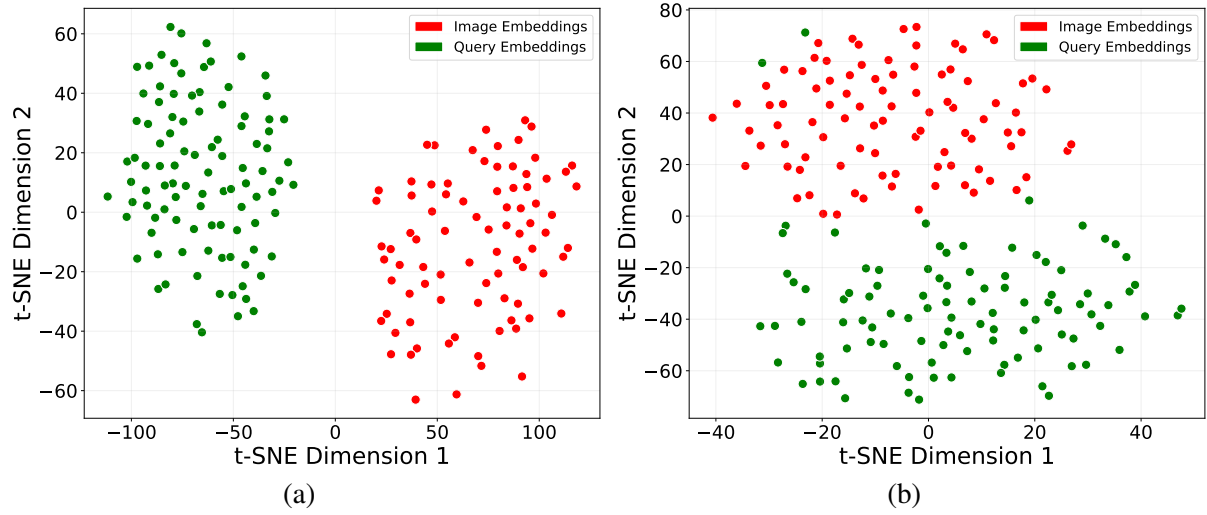


Figure 5: t-SNE visualizations of embeddings generated by (a) CLIP-L and (b) Jina-CLIP.

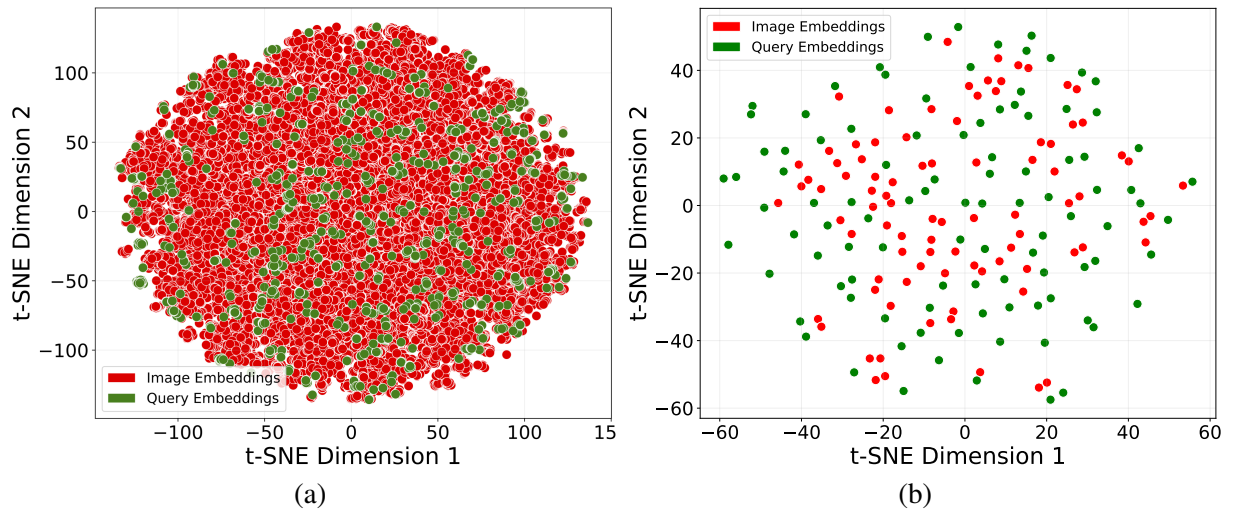


Figure 6: t-SNE visualizations of (a) all embeddings and (b) average embeddings per image-query pair generated by ColPali.